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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.17

$$\int \frac{6x^2+7x+5}{3x^2+7x+2}dx$$

$$3x^2+7x+2=(3x+1)(x+2)$$

$$\frac{6x^2+7x+5}{3x^2+7x+2}=2+\frac{-7x+1}{3x^2+7x+2}$$

$$\frac{-7x+1}{3x^2+7x+2}=\frac{-7x+1}{(3x+1)(x+2)}=\frac{A}{3x+1}+\frac{B}{x+2}$$

$$A=\left.\frac{-7x+1}{x+2}\right|_{x=-\frac{1}{3}}=\frac{\frac{10}{3}}{\frac{5}{3}}=2$$

$$B=\left.\frac{-7x+1}{3x+1}\right|_{x=-2}=\frac{15}{-5}=-3$$

$$\frac{6x^2+7x+5}{3x^2+7x+2}=2+\frac{2}{3x+1}-\frac{3}{x+2}$$

$$\int \frac{6x^2+7x+5}{3x^2+7x+2}dx=\int 2dx+2\int \frac{dx}{3x+1}-3\int \frac{dx}{x+2}$$

$$=2x+\frac{2}{3}\ln|3x+1|-3\ln|x+2|+C$$

References